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# ICCAD 2026 CAD Contest Problem D

## Timing Fixing by Useful Skew

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### Abstract

## 1 Introduction

ICCAD Contest 2026 Problem D asks us to improve such violations by modifying the clock tree rather than changing the data-path logic. The available operations are local clock-tree edits: inserting a new buffer, removing an inserted buffer, or resizing an existing buffer. These operations change clock arrival times at flip-flops and therefore change the skew between launch and capture registers.

The key opportunity is useful skew. A nonzero clock skew is not always harmful: delaying the capture clock can give a setup-critical data path more time, while delaying the launch clock can help avoid hold violations by postponing data launch. However, each clock-tree edit may affect many data paths at once. Improving one endpoint may degrade another path, and adding buffers also increases area. The optimization problem is therefore not only to remove timing violations, but to balance setup improvement, hold improvement, and area cost.

In this project, we model the clock tree as a mutable state and search over a space of legal buffer insertion, removal, and resizing operations. Our solver uses candidate policies to decide which moves should be evaluated, and acceptance policies to decide which evaluated move should be applied. This framework allows us to compare greedy search, simulated annealing, iterated simulated annealing, two-step timing repair, and tabu search under a common clock-tree timing model. The experimental discussion focuses on how these policies trade off setup slack, hold slack, and area.

## 2 Problem Formulation

### 2.1 Slack, Skew, and Score

A data path connects a launch flip-flop to a capture flip-flop through combinational logic. Let  $D$  be the data-path delay,  $T_{\text{clk}}$  be the clock period,  $T_{\text{setup}}$  be the setup requirement, and  $T_{\text{hold}}$  be the hold requirement. Let  $C_{\text{launch}}$  and  $C_{\text{capture}}$  denote the clock arrival times from the clock source to the launch and capture flip-flops. We define clock skew as

$$\text{skew} = C_{\text{capture}} - C_{\text{launch}}. \quad (1)$$

With this convention, the setup and hold slacks used in the presentation are

$$\text{setup\_slack} = T_{\text{clk}} - T_{\text{setup}} - D + \text{skew}, \quad (2)$$

$$\text{hold\_slack} = D - T_{\text{hold}} - \text{skew}. \quad (3)$$

A negative setup slack means that data arrives too late at the capture flip-flop. A negative hold slack means that data arrives too early and may overwrite the previous value. Therefore, a setup violation can often be improved by increasing the capture clock delay, while a hold violation can often be improved by increasing the launch clock delay.

The objective combines timing and area. Timing quality is measured by total negative slack (TNS) and worst negative slack (WNS) under the setup-oriented SS corner and the hold-oriented FF corner. Area is the total area of the clock-tree

buffers. Given a baseline metric value  $x_0$  and a final value  $x$ , we use the relative improvement form

$$I(x, x_0) = 1 - \frac{x}{x_0}, \quad (4)$$

when  $x_0$  is nonzero. This rewards reducing negative timing metrics toward zero and rewards reducing area below the baseline. The implemented score is

$$\begin{aligned} S_{\text{setup}} &= I(\text{TNS}_{SS}, \text{TNS}_{SS}^0) + I(\text{WNS}_{SS}, \text{WNS}_{SS}^0), \\ S_{\text{hold}} &= I(\text{TNS}_{FF}, \text{TNS}_{FF}^0) + I(\text{WNS}_{FF}, \text{WNS}_{FF}^0), \\ S_{\text{area}} &= 1 - \frac{\text{Area}}{\text{Area}^0}, \\ S &= \alpha S_{\text{setup}} + \beta S_{\text{hold}} + \gamma S_{\text{area}}, \\ \alpha &= 0.5, \quad \beta = 0.25, \quad \gamma = 0.25. \end{aligned} \quad (5)$$

This formula explains the main trade-off in the problem: aggressive buffer insertion may improve setup or hold slack, but the same move can reduce the area component and may also hurt timing on other paths. The actual parameter is not given; therefore, we set parameters based on the importance for evaluation.

## 2.2 Optimization Moves

The state space consists of all clock trees reachable from the input tree by legal edits. A state records which buffers are present, the cell type of each buffer, and the parent-child connectivity of the clock tree. Original contest nodes remain part of the design, while buffers inserted by the optimizer may be removed later.

Our optimizer uses three move types. First, an *insert* move places a new buffer on an existing clock-tree edge. This increases the clock delay of the downstream subtree and can be used to adjust useful skew, but it also increases area. Second, a *remove* move deletes a buffer that was inserted by the optimizer and reconnects the surrounding tree. This can recover area, but it may undo a previous timing repair. Third, a *resize* move changes the cell type of an existing buffer. Resizing gives a finer way to increase or decrease clock delay and area without changing the clock-tree topology.

Because the number of possible edit sequences is large, the solver cannot enumerate the full state space. Instead, each iteration generates a smaller set of candidate moves and evaluates their impact on the current timing and score. The rest of the report describes how we construct these candidate sets and how different search policies decide which move to accept.

## 3 Methodology

### 3.1 Model for Optimization

**Clock Tree.** The clock tree is the primary data structure in our optimization framework. It consists of the clock source as the root, buffers as internal nodes, and flip-flops as leaves. The supported operations include buffer insertion, buffer removal, and buffer resizing.

**Data Path Graph.** The data path graph stores the launch flip-flop, capture flip-flop, and data-path delay information. This information is required to evaluate whether a clock-tree modification improves timing slack.

**Timing State.** The timing state maintains timing and area metrics, including setup slack, hold slack, total negative slack (TNS), worst negative slack (WNS), and area. These metrics are updated and queried during optimization.

#### Design Philosophy.

*Minimize Re-computation.* Recomputing timing information for the entire clock tree after every modification is computationally expensive. Therefore, we employ an incremental timing cache that updates only the affected paths instead of the entire structure.

*Recoverability.* The optimization process must support undo operations so that a modification can be reverted efficiently when the move is rejected.

### 3.2 Candidate Policies

**Violation Path.** We check all data paths and find out those whose slack is negative and smallest, which means the data path violate timing constrain. If setup constrain is violated, the buffers near capture flip-flop is modified. Otherwise, if hold constrain is violated, the buffers near launch flip-flop is modified

**Upstream Window.** We look for violation endpoint (i.e. capture flip-flop) and explore several buffers toward upstream of clock tree. The policy adds more candidate for optimization, but also more time consuming.

**Critical Endpoint.** We define *criticality* as an evaluation metric for how much a flip-flop worsen total negative slack. Negative setup slack is added to violation path’s capture flip-flop’s criticality, and negative hold slack is added to launch flip-flop’s criticality, respectively. We try to modify the flip-flop with largest criticality to optimize the violation on more than one data path.

**Random.** The policy randomly generate a set of moves from inserting buffer, removing inserted buffer, and resizing buffer with a certain probability.

### 3.3 Accept Policies

**Greedy Approach.** Based on chosen candidate policy, we apply all generated candidate operations to current clock tree and calculate  $\Delta Score$ . The strategy accept only operations with  $\Delta Score > 0$  and the largest  $\Delta Score$ .

**Two Step Optimization.** It’s similar to greedy approach but with two stage focusing on different objective. In the first stage, we try to optimize timing constrain and neglect the penalty caused by area increment. In the second stage, we try to remove and resize buffers to reduce area penalty with timing performance maintained.

**Simulated Annealing (SA).** The strategy randomly come up with an move from candidate pool and apply the move on current clock tree. If  $\Delta Score > 0$  then the move is accepted, otherwise with a certain probability determined by the temperature, this move will be accepted. When the temperature is low, the strategy act like greedy approach.

**Iterated Simulated Annealing (ISA).** The strategy take turns doing standard simulated annealing step and greedy step, trying to explore more possible moves.

**Tabu Search.** In each step of the strategy, we pick the best move among all candidates greedily. If the move is non-tabu move, then we directly accept the move, otherwise only if the move creates new global optimal will the move be accepted.

**Candidate Pool.** In our design of simulated annealing and tabu search, the candidate pool is either *union pool* or *sampled union pool*. The union pool explicitly collects candidates generated by different candidate policies and compares them together. In contrast, sampled union pool first samples one candidate family and then samples one move from that family. This keeps each stochastic proposal cheap, but it may miss the best move in the full union pool at that iteration.

## 4 Experimental Results and Discussion

### 4.1 Using Different Candidate Policy

We first compare different candidate policies under the greedy algorithm.

**Score Analysis.** Overall, union pool is the best candidate policy, while random and upstream window are close behind. Violation path and critical endpoint perform worse because they do not improve both setup and hold slack sufficiently. The stronger policies have similar setup performance, but union pool gains extra score from better hold improvement and more controlled area.

**Possible Reason.** Greedy search always chooses the best immediately improving move. Therefore, its performance depends heavily on whether the candidate pool contains a good local edit. Violation path and critical endpoint generate narrower candidate sets, so they are more likely to miss useful upstream or area-recovery moves.

### 4.2 Using Different Accept Policy

We next compare accept policies using random candidates, union pool, and sampled union pool.

#### 4.2.1 Random Candidate

With random candidates, tabu search gives the best result, while two-step optimization is the weakest. The main issue of two-step optimization is that its first stage focuses on timing repair and its second stage tries to recover area afterward. This separation is too rigid: area-recovery moves may undo previous timing repair, while timing-focused moves may leave too much area cost for the second stage to clean up. As a result, two-step optimization has weaker hold slack and area than SA, ISA, and tabu search. Its timing-first stage also does not outperform SA/ISA enough to justify the later cleanup cost. Tabu search performs better because it can accept non-improving intermediate moves, avoid recent cycles with tabu memory, and still restore the best solution found.

#### 4.2.2 Union Pool and Sampled Union Pool

With union pool and sampled union pool, two-step optimization is still the weakest. Although union pool provides more candidates, the two fixed stages do not convert this larger search space into a better final score. SA and ISA aggressively improve setup slack, but they pay a large area penalty because many accepted moves insert or preserve extra buffers. Tabu search is more balanced: it improves timing while keeping the final area close to the baseline.

### 4.3 Overall Performance

Table 1 reports the average over five testcases and five seeds. Area ratio is the final area divided by the baseline area, so values below one indicate area reduction. Std. is the standard deviation of the final scores over these runs.

Table 1: Performance comparison of different optimizers.

| Optimizer                | Avg. Score | Setup     | Hold      | Area       | Area Ratio | Std.      |
|--------------------------|------------|-----------|-----------|------------|------------|-----------|
| tabu-random              | 1.140 722  | 1.276 422 | 1.916 255 | 0.093 788  | 0.906 212  | 0.235 526 |
| isa-random               | 1.084 773  | 1.322 733 | 1.737 303 | -0.043 676 | 1.043 676  | 0.273 375 |
| tabu-union-pool          | 1.062 058  | 1.259 221 | 1.731 261 | -0.001 474 | 1.001 474  | 0.274 410 |
| isa-sampled-union-pool   | 1.044 759  | 1.521 136 | 1.721 173 | -0.584 406 | 1.584 406  | 0.183 581 |
| sa-sampled-union-pool    | 1.044 015  | 1.653 149 | 1.833 949 | -0.964 191 | 1.964 191  | 0.165 838 |
| greedy-union-pool        | 1.042 167  | 1.193 238 | 1.757 868 | 0.024 327  | 0.975 673  | 0.266 567 |
| greedy-upstream-window   | 1.031 489  | 1.227 970 | 1.710 767 | -0.040 751 | 1.040 751  | 0.275 511 |
| greedy-random            | 1.009 059  | 1.104 560 | 1.806 695 | 0.020 420  | 0.979 580  | 0.214 807 |
| sa-random                | 0.998 808  | 1.149 864 | 1.750 100 | -0.054 596 | 1.054 596  | 0.276 991 |
| two-step-random          | 0.876 061  | 1.148 869 | 1.334 986 | -0.128 479 | 1.128 479  | 0.318 905 |
| two-step-union-pool      | 0.859 649  | 1.045 729 | 1.405 060 | -0.057 920 | 1.057 920  | 0.401 216 |
| greedy-violation-path    | 0.829 324  | 1.032 678 | 1.300 464 | -0.048 521 | 1.048 521  | 0.277 805 |
| greedy-critical-endpoint | 0.746 583  | 0.923 081 | 1.188 662 | -0.048 491 | 1.048 491  | 0.375 666 |

The table shows that maximizing one timing component is not sufficient. For example, *sa-sampled-union-pool* has the highest setup component, but its area ratio is almost two times the baseline. This area penalty lowers its final score below tabu-based methods. In contrast, *tabu-random* has a lower setup component, but it achieves the best hold component and reduces area, which leads to the best overall score.

### 4.4 Difference Between Random and (Sampled) Union Pool on ISA and Tabu

The best strategy is tabu search with random candidates. ISA with sampled union pool improves setup slack the most, but this improvement is obtained at the cost of much larger area. In contrast, tabu search with random candidates still improves setup slack significantly, achieves the strongest hold improvement, and reduces area consumption. Therefore, its advantage comes from the best trade-off among setup, hold, and area rather than from maximizing a single component.

#### 4.4.1 Possible Reason

ISA and tabu search are both designed to escape local optima. Random candidates provide wider exploration, while tabu memory prevents the search from repeatedly returning to recently used moves. This also explains why *tabu-random* outperforms *tabu-union-pool*: union pool is more timing-aware, but it can bias the search toward violation-related edits, while random candidates expose more non-local resize and removal opportunities. This combination works better than a purely timing-focused sampled union proposal when area must also be controlled.

## 5 Conclusion

In this work, we formulate timing fixing by useful skew as a clock-tree optimization problem over three reversible operations: buffer insertion, inserted buffer removal, and buffer resizing. These operations allow the optimizer to change clock arrival times at flip-flops and improve setup or hold slack without modifying the data-path logic. Since each edit may affect multiple paths and also changes area, the main challenge is not only timing repair, but the balance among setup slack, hold slack, and area cost.

Our experiments show that both candidate generation and acceptance strategy have a significant impact on the final result. Greedy search benefits from a larger candidate pool because it only accepts immediately improving moves. In contrast, simulated annealing and tabu search can explore beyond local optima, which makes random candidate generation more useful. Among all tested optimizers, tabu search with random candidates achieves the best average score. It does not produce the highest setup component, but it achieves the strongest hold improvement while also reducing area, giving the best overall trade-off. SA and ISA can aggressively improve timing, especially setup slack, but their area penalty reduces the final score. Two-step optimization follows a reasonable timing-first and area-recovery idea, but its fixed phase separation does not recover enough hold slack and area to compete with tabu search.

For future work, an adaptive strategy could select candidate policies according to the current violation pattern instead of using a fixed policy throughout the run. Another promising direction is to keep the exploration strength of tabu search while adding a stronger area-aware resize and removal polish in the late stage. This may preserve the timing gains from useful skew while further reducing unnecessary buffer area.

## **Author Contributions**

Hong Bin Lai did nothing :)

## **Generative AI Usage Disclosure**

We used generative AI tools during the development of this project. In particular, OpenAI Codex and Cursor was used extensively to assist with source-code implementation, including parser construction, clock-tree data structures, timing evaluation routines, optimization-loop implementation, test scripts, and refactoring. Approximately 80% of the submitted source code was initially generated or modified with the assistance of Codex or Cursor.

The algorithmic ideas, including the useful-skew optimization strategy, candidate generation policies, acceptance strategies, timing-area trade-off considerations, experimental design, and final optimizer selection, were designed and evaluated by the team members. All AI-generated code was manually reviewed, integrated, debugged, and tested by the team. The reported experimental results were obtained by executing our implemented solver on the test cases; no experimental data or performance numbers were generated by AI tools.

We also used ChatGPT/Codex to assist with report organization, wording refinement, and documentation polishing. The final technical content, claims, results, and conclusions were checked and approved by the team.

## **References**